Pin Out Table

Stepper Carriage Positioning — Micro850 PLC | 24 VDC System

Pin #	Group	Device / Signal	Wire Function	Voltage
1	Stepper Motor	Stepper Motor (STP-MTR-23055, NEMA 23)	A+ (RED) → CW250 A+	—
2	Stepper Motor	Stepper Motor (STP-MTR-23055, NEMA 23)	A– (WHT) → CW250 A–	—
3	Stepper Motor	Stepper Motor (STP-MTR-23055, NEMA 23)	B+ (GRN) → CW250 B+	—
4	Stepper Motor	Stepper Motor (STP-MTR-23055, NEMA 23)	B– (BLK) → CW250 B–	—
5	Stepper Driver	Stepper Driver (CW250/2/8)	VCC ← Driver power supply +	20–60 VDC
6	Stepper Driver	Stepper Driver (CW250/2/8)	GND ← Driver power supply –	Driver 0 V
7	Stepper Driver	Stepper Driver (CW250/2/8)	CP+ (pulse) ← PLC DO_01 (PTO)	24 VDC
8	Stepper Driver	Stepper Driver (CW250/2/8)	CP– → –24 VDC common	–24 VDC
9	Stepper Driver	Stepper Driver (CW250/2/8)	CW+ (direction) ← PLC PTO direction output	24 VDC
10	Stepper Driver	Stepper Driver (CW250/2/8)	CW– → –24 VDC common	–24 VDC
11	Stepper Driver	Stepper Driver (CW250/2/8)	H/D (½ / ⅛ step select) ← PLC DO_06	24 VDC
12	Green Pushbutton	Pushbutton, Green, 1 N.O. (GCX1102) — Left	N.O. contact → PLC DI_07	24 VDC
13	Green Pushbutton	Pushbutton, Green, 1 N.O. (GCX1102) — Left	N.O. contact ← +24 VDC	+24 VDC
14	Red Pushbutton	Pushbutton, Red (GCX1101 + ECX1040 N.O.) — Right	N.O. contact → PLC DI_02	24 VDC
15	Red Pushbutton	Pushbutton, Red (GCX1101 + ECX1040 N.O.) — Right	N.O. contact ← +24 VDC	+24 VDC
16	Yellow Pushbutton	Pushbutton, Yellow, 1 N.O. (GCX1103) — Center	N.O. contact → PLC DI_04	24 VDC
17	Yellow Pushbutton	Pushbutton, Yellow, 1 N.O. (GCX1103) — Center	N.O. contact ← +24 VDC	+24 VDC
18	Enable Switch	Selector, 2-pos maintained, 1 N.O. (GCX1350) — Enable	N.O. contact → PLC DI_03	24 VDC
19	Enable Switch	Selector, 2-pos maintained, 1 N.O. (GCX1350) — Enable	N.O. contact ← +24 VDC	+24 VDC
20	Limit Switch 1	Micro Limit Switch (WS1-Z5-W300R300) — Left / Home	COM ← +24 VDC	+24 VDC
21	Limit Switch 1	Micro Limit Switch (WS1-Z5-W300R300) — Left / Home	NO → PLC DI_05	24 VDC
22	Limit Switch 1	Micro Limit Switch (WS1-Z5-W300R300) — Left / Home	NC — not used (insulate)	—
23	Limit Switch 2	Micro Limit Switch (WS1-Z5-W300R300) — Right	COM ← +24 VDC	+24 VDC
24	Limit Switch 2	Micro Limit Switch (WS1-Z5-W300R300) — Right	NO → PLC DI_06	24 VDC
25	Limit Switch 2	Micro Limit Switch (WS1-Z5-W300R300) — Right	NC — not used (insulate)	—
26	Proximity Sensor 1	Inductive Prox. (PNM6-AP-3A) — Left-most	Signal wire (BK) → PLC DI_09	24 VDC
27	Proximity Sensor 1	Inductive Prox. (PNM6-AP-3A) — Left-most	Power –24 VDC (BU)	–24 VDC
28	Proximity Sensor 1	Inductive Prox. (PNM6-AP-3A) — Left-most	Power +24 VDC (BN)	+24 VDC
29	Proximity Sensor 2	Inductive Prox. (PNM6-AP-3A) — Center	Signal wire (BK) → PLC DI_08	24 VDC
30	Proximity Sensor 2	Inductive Prox. (PNM6-AP-3A) — Center	Power –24 VDC (BU)	–24 VDC
31	Proximity Sensor 2	Inductive Prox. (PNM6-AP-3A) — Center	Power +24 VDC (BN)	+24 VDC
32	Proximity Sensor 3	Inductive Prox. (PNM6-AP-3A) — Right-most	Signal wire (BK) → PLC DI_10	24 VDC
33	Proximity Sensor 3	Inductive Prox. (PNM6-AP-3A) — Right-most	Power –24 VDC (BU)	–24 VDC
34	Proximity Sensor 3	Inductive Prox. (PNM6-AP-3A) — Right-most	Power +24 VDC (BN)	+24 VDC
35	Green Status Light	LED Pilot Light, Green (GCX1232-24L) — Left	Lamp + ← PLC DO_07	+24 VDC
36	Green Status Light	LED Pilot Light, Green (GCX1232-24L) — Left	Lamp – → –24 VDC common	–24 VDC
37	Red Status Light	LED Pilot Light, Red (GCX1231-24L) — Right	Lamp + ← PLC DO_08	+24 VDC
38	Red Status Light	LED Pilot Light, Red (GCX1231-24L) — Right	Lamp – → –24 VDC common	–24 VDC
39	Yellow Status Light	LED Pilot Light, Yellow (GCX1233-24L) — Center	Lamp + ← PLC DO_09	+24 VDC
40	Yellow Status Light	LED Pilot Light, Yellow (GCX1233-24L) — Center	Lamp – → –24 VDC common	–24 VDC
41	HMI Display	PanelView 800 4.3" TFT HMI (2711R-T4T)	Power +24 VDC	+24 VDC
42	HMI Display	PanelView 800 4.3" TFT HMI (2711R-T4T)	Power –24 VDC	–24 VDC
43	HMI Display	PanelView 800 4.3" TFT HMI (2711R-T4T)	Ethernet ↔ Micro850 Ethernet port	—
44	PLC Power & Commons	Allen-Bradley Micro850 (2080-LC50-24QBB)	+24 VDC ← PSB24-060S +V	+24 VDC
45	PLC Power & Commons	Allen-Bradley Micro850 (2080-LC50-24QBB)	–24 VDC ← PSB24-060S –V	–24 VDC
46	PLC Power & Commons	Allen-Bradley Micro850 (2080-LC50-24QBB)	DI COM → –24 VDC (sinking inputs for PNP devices)	–24 VDC
47	PLC Power & Commons	Allen-Bradley Micro850 (2080-LC50-24QBB)	DO +CM ← +24 VDC	+24 VDC
48	PLC Power & Commons	Allen-Bradley Micro850 (2080-LC50-24QBB)	DO –CM → –24 VDC	–24 VDC

Color Legend
Stepper Motor — STP-MTR-23055 (coil leads A± / B±)
Stepper Driver — CW250/2/8 (power, CP / CW pulse-direction, H/D)
Operator Controls (GCX 22 mm pushbuttons, enable selector)
Limit Switches 1 & 2 — WS1-Z5-W300R300 (left / home, right)
Proximity Sensor 1 — Left-most
Proximity Sensor 2 — Center
Proximity Sensor 3 — Right-most
Status Lights — GCX 24 V LED pilot lights
PanelView 800 4.3" TFT touch screen HMI
Micro850 PLC power and I/O commons

Notes
CW250 inputs: confirm CP / CW / H/D accept 24 V signals; if rated 5 V, add a series resistor per the driver manual.
Set CW250 phase current to the motor rating (2.8 A/phase for STP-MTR-23055). H/D assumed to select ½ / ⅛ step ("/2/8").
PTO direction output is set in CCW Motion › X_axis; wire CW+ to that output.
GCX1101 ships with 1 N.C.; fit an ECX1040 1 N.O. block so the red button matches the program.
Lead colors from the motor and sensor datasheets; confirm against the parts received.

Stepper Carriage BOM				
#	Description	Part Number	Quantity	Source
1	Micro850 PLC, 24 VDC, 14 in / 10 out, PTO	2080-LC50-24QBB	1	Rockwell (Allen-Bradley)
2	Stepper driver, 20–60 VDC, 5 A, ½ / ⅛ step	CW250/2/8	1	RicoCNC (cncsparetools.com)
3	Stepper motor, NEMA 23, 2.8 A/phase, 166 oz·in	STP-MTR-23055	1	AutomationDirect
4	Pushbutton 22 mm, green, flush, 1 N.O.	GCX1102	1	AutomationDirect
5	Pushbutton 22 mm, red, flush (ships 1 N.C.)	GCX1101	1	AutomationDirect
6	Contact block 22 mm, 1 N.O. (for red pushbutton)	ECX1040	1	AutomationDirect
7	Pushbutton 22 mm, yellow, flush, 1 N.O.	GCX1103	1	AutomationDirect
8	Selector switch 22 mm, 2-pos maintained, 1 N.O.	GCX1350	1	AutomationDirect
9	Micro limit switch, IP67, SPDT, arc roller lever, 300 mm leads	WS1-Z5-W300R300	2	Bituo Electric (via Amazon)
10	Inductive prox, 12 mm, PNP N.O., 4 mm flush, 10–30 VDC	PNM6-AP-3A	3	AutomationDirect
11	LED pilot light 22 mm, green, 24 VAC/VDC	GCX1232-24L	1	AutomationDirect
12	LED pilot light 22 mm, red, 24 VAC/VDC	GCX1231-24L	1	AutomationDirect
13	LED pilot light 22 mm, yellow, 24 VAC/VDC	GCX1233-24L	1	AutomationDirect
14	PanelView 800 HMI, 4.3" TFT touch, 24 VDC	2711R-T4T	1	Rockwell (Allen-Bradley)
15	Power supply, 24 VDC, 2.5 A, 60 W, DIN rail	PSB24-060S	1	AutomationDirect
16	DIN terminal block, feed-through, gray, 24–10 AWG (pack of 100)	DN-T10-A	1	AutomationDirect
17	End cover for DN-T10-A (pack of 100)	DN-EC1210	1	AutomationDirect
		Total qty	**20**	

Part numbers link to the datasheet or listing used.